

IT694
COMPUTER NETWORKS
MID SEM I SOLUTIONS, WINTER 2025

1. Consider sending a packet from a source host to a destination host over a fixed route. List the delay components in the end-to-end delay. Which of these delays are constant and which are variable? (0.5 each)

SOLN:

- Transmit delay – Variable
- Propagation delay – Constant
- Queuing delay - Variable
- Processing delay – constant

2. Suppose A has a TCP connection with B, where A sends data segments and B sends ACKs for every segment received from A. Let the Maximum Segment Size (MSS) be 472 bytes, and all packets sent by A have this size. Assume that all packets have 20 bytes of TCP, IP and data link-layer header. Assume the combined data and ACK packets fully utilize the middle link in both directions. What fraction of the bandwidth is consumed by data traffic?

SOLN:

- Total header size (The link-layer header 20 bytes + TCP header 20 bytes + IP header 20 bytes = 60 bytes of header. – **[1 point]**)
 - An ACK packet contains no payload (i.e., no segment) and, as such, is 60-bytes long.
 - A full-sized TCP data packet includes MSS bytes (472 bytes + 60-byte header = 532-byte data packets.
 - So, sending one TCP segment (472 bytes) includes ONE data packet of 532 bytes and an ACK of 60 bytes
 - This leads to a ratio of 472/592 **[0.5 point]**
 - 79.7% of the link bandwidth. **[0.5 point]**
3. Using TCP, a sender has sent out a total of ten data segments, each of 1000 bytes. Assume that the sender's initial sequence number (ISN) is 5000. The sender receives two more packets with "ACK 8001". Given this occurrence, which of the TCP segments does the sender conclude might be lost and thus should be retransmitted?

SOLN:

It may conclude packet 8001-9000 is lost and should be retransmitted

4. Using TCP, a sender has sent out a total of ten data segments, each of 1000 bytes. Assume that the sender's initial sequence number (ISN) is 5000. After the sender receives a packet with "ACK 8001", how many TCP segments, if any, does the sender know that the receiver definitely received?

SOLN:

3 packets (6000, 7000, 8000)

5. State and justify in one line if True or False: When this line `socket.accept()` is executed, it returns a socket associated with a new TCP connection. [1 point]

Yes **[0.5 points]**

When the `socket.accept()` function is executed, the Server accepts a new TCP connection with a client [0.5 points]

6. Say Yes or No and justify in one line: When this function `socket.accept()` is executed, it returns 'true' if a new TCP connection request has arrived, false otherwise. [1 point]
SOLN:

No [0.5 points]

When the `socket.accept()` function is executed, the Server accepts a new TCP connection with a client [0.5 points]

7. ATM-Server protocol problem:

If the answers are similar to the below, give full marks [4 points]

Messages from ATM machine to Server

Msg name	purpose
-----	-----
HELO <userid>	Let server know that there is a card in the ATM machine ATM card transmits user ID to Server
PASSWD <passwd>	User enters PIN, which is sent to server
BALANCE	User requests balance
WITHDRAWL <amount>	User asks to withdraw money
BYE	user all done

Messages from Server to ATM machine (display)

Msg name	purpose
-----	-----
PASSWD	Ask user for PIN (password)
OK	last requested operation (PASSWD, WITHDRAWL) OK
ERR	last requested operation (PASSWD, WITHDRAWL) in ERROR
AMOUNT <amt>	sent in response to BALANCE request
BYE	user done, display welcome screen at ATM

- b) Diagram: Again if the protocol steps are similar, give max marks

Correct operation:

client	server
HELO (userid) ----->	(check if valid userid)
<-----	PASSWD
PASSWD <passwd>	-----> (check password)
<-----	OK (password is OK)
BALANCE ----->	
<-----	AMOUNT <amt>
WITHDRAWL <amt>	-----> check if enough \$ to cover withdrawl
<-----	OK
ATM dispenses \$	
BYE ----->	
<-----	BYE

In situation when there's not enough money:

HELO (userid) ----->	(check if valid userid)
<-----	PASSWD
PASSWD <passwd>	-----> (check password)
<-----	OK (password is OK)

```
BALANCE      ----->
              <-----
WITHDRAWAL <amt>
              <-----
error msg displayed
no $ given out
BYE          ----->
              <-----  AMOUNT <amt>
                          ----->  check if enough $ to cover withdrawl
                          ERR (not enough funds)
BYE
```